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ANT COLONY OPTIMIZATION ALGORITHM FOR MULTI-DEPOT VEHICLE ROUTING PROBLEM WITH TIME WINDOWS

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Abstract. *The Multi-Depot Vehicle Routing Problem with Time Windows (MDVRP-TW) is a well-known optimization problem with many practical applications in a wide range of domains. The primary goal is to find optimal routes for a set of vehicles starting from multiple depots to a number of customers. Each customer should be served only once and each vehicle returns to the original depot after visiting all customers along its route. Each vehicle can visit a customer only during its particular predefined time window. Also the condition of maximum vehicles' load has to be met. There are many heuristic and metaheuristic algorithms proposed for this problem solution (e.g. tabu search, simulated annealing, genetic algorithms) as this is a NP-hard problem and, therefore, exact methods are not feasible for more complex solutions. Another possibility is to apply the Ant Colony Optimization (ACO) theory to this problem. The authors successfully managed to adapt the ACO algorithm for the MDVRP-TW problem. The algorithm has been verified on a set of Cordeau's test instances. In all tested cases, the solution found by our algorithm compared with the best solution known so far is not worse than 8%. In addition, the great advantage is that the algorithm is very fast as the key processes can be executed in parallel (e.g. on multiple cores of a processor). The article presents an original solution of authors to the MDVRP-TW problem via the ACO algorithm in detail. The first part deals with the algorithm including its principles and parameters. Then several examples and experiments are shown. The last part of the article shows conclusions and future work of the authors.*

1 INTRODUCTION

The Multi-Depot Vehicle Routing Problem with Time Windows (MDVRP-TW) is a well-known problem with many real applications in the areas of transportation, distribution and logistics. It is an extension of a classic MDVRP problem formulated by Dantzig and Ramser in 1959 [1]. The difference consists in a predefined time interval in which a customer has to be served.

There are a lot of existing algorithms used successfully to solve the problem. These can be classified as (a) exact, (b) heuristic, and (c) metaheuristic. Especially the latter have underwent the fast development recently, as this is an NP-hard problem and exact methods are not often feasible.

The primary objective of this paper is to introduce the new original approach of authors to this problem based on applying Ant Colony Optimization (ACO) theory. It is a nature-inspired idea introduced in 1992 by Dorigo [2]; he proposed the algorithm to search for the optimal path in a graph.

Section 2 introduces the problem formulation and literature review. In Section 3, we present the key principles and parameters of our algorithm. Section 4 shows experiments we conducted along with the results obtained on a set of Cordeau's test instances [3]. Finally in Section 5, we show same perspectives of our future work.

2 MULTI-DEPOT VEHICLE ROUTING PROBLEM WITH TIME WINDOWS

This section covers a formal definition of the problem and the review of some state of the art approaches proposed for solution.

2.1 Problem formulation

Let $G = (V, E)$ be a graph where $V = S \cup D$ is a set of customers $S = \{S_1, S_2, \dots, S_N\}$ and depots $D = \{D_1, D_2, \dots, D_M\}$ and E is a set of edges between all nodes in the graph. Each depot D_i can have one or more vehicles at its disposal $K_i = \{k_{i1}, k_{i2}, \dots\}$. The total set of vehicles $K = K_1 \cup K_2 \cup \dots \cup K_M$.

For every edge E_{ij} between existing nodes V_i and V_j $(i \neq j)$, a non-negative cost c_{ij} is associated which can be interpreted as a travel cost or as a travel time. Each vehicle with capacity Q_k $(k \in K)$ leaves from its depot with the purpose of collecting (or distributing respectively) a certain quantity $q_i (i \in S)$ from (or to) a sequence of customers providing that the capacity of the vehicle is not exceeded $(\sum_i q_i \leq Q_k)$. After the trip, the vehicle has to return to its depot. Formula (1) presents the main objective which is to minimize the total cost spent by all vehicles.

$$\min \sum_{i \in V} \sum_{j \in V} \sum_{k \in K} c_{ij} \cdot x_{ij}^k, \qquad \text{where } x_{ij}^k = \begin{cases} 1 & \text{if } i \text{ preceeds } j \text{ on route } k, \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

There are some limiting constraints: the delivery has to be done within a customer-specified time window; each customer has to be served only once; load of each vehicle do not have to exceed its capacity.

2.2 Literature review

As already mentioned, different approaches to solve the MDVRP-TW problem exist. When taking into account only the classic MDVRP problem, a broad overview of various methods

can be found e.g. in [4], [5]. In particular, very popular methods for this problem are introduced e.g. [6], [7], [8], to name a few.

Methods for the MDVRP-TW problem have been inspired in many cases by methods for the simpler MDVRP problem. Several examples of such algorithms can be found in [9], [10], [11]. In addition, very popular metaheuristic methods have emerged in a last few years, for instance genetic algorithms [12].

As the topic of this article deals with using the ACO theory to solve the problem, the remaining part of this section presents the publications concerning this domain. The potential of the ACO principle was detected very soon after it had been discovered and formulated. Recently, some articles have emerged using the ACO theory to solve the MDVRP problem, see [13], [14], [15].

There has not been published a solution to the MDVRP-TW problem based entirely on the ACO theory yet (at least as the authors are aware of). However, some attempts to use this theory in combination with another principle exist, e.g. Ting and Chen [16] published the paper dealing with the combination of the ACO theory and simulated annealing to solve the MDVRP-TW problem.

3 ANT COLONY OPTIMIZATION ALGORITHM

ACO algorithm is a probabilistic technique for solving computational problems. It is a member of swarm intelligence methods. The principle is adopted from the natural world where ants explore the environment to find food. The exploring starts randomly at first; when an ant finds food, however, it lays down a pheromone trail along its route. When other ants find such a trail, they are likely to follow it (and lay down their own trail if successful). Pheromone trails evaporate gradually, thus reducing its strength; is has an effect of avoiding the convergence to a locally optimal solution.

3.1 Principle

Figure 1 presents the ACO algorithm we proposed for the MDVRP-TW problem. The solution found by the algorithm is being improved steadily in successive generations (iterations). In point 1, the termination condition is tested, points 2 to 19 covers an individual generation. Each depot represents a colony with the specific number of ants.

In each generation, all ants in colonies wander between individual customers (referred to as nodes in Figure 1). At first, state of all nodes is set as unvisited in point 3. Then, the algorithm continues until all nodes are visited (just once). In point 5, the algorithm checks whether the feasible solution still exists, i.e. whether there are still nodes which can be visited by any depot during their predefined time interval. If not, the algorithm starts from the beginning. In point 7, the depot (colony) is selected according to the given method; points 8 to 15 then apply only to the ant from the selected colony.

Next, ant's probability is computed; it determines the chance of the ant to go to every remaining unvisited nodes. The process also reflects the condition of visiting nodes only during their predefined time intervals. In case it is impossible to visit the node by the ant in time, the node is assigned zero probability. If there is no node to be selected (see points 9), the algorithm proceeds to point 5 and attempts to select another depot. In point 11, a node is chosen according to the computed probabilities.

Point 12 checks whether the ant can visit the selected node (i.e. whether its current load allows taking the load in the node and thus not exceeding ant's maximum capacity). If not, the ant returns to its colony (emptying its load) and the algorithm continues in point 4. If yes on the other hand, the ant visits the selected node (taking node's load and marking it as visited).

```
1.  while not terminated
2.    for each ant in a colony
3.      set all nodes as unvisited
4.      while number of unvisited nodes > 0 do
5.        if there is no feasible solution then
6.          go to 1
7.        select a depot
8.          compute ant's probability of going to unvisited nodes
9.          if there is no node to be selected then
10.           go to 5
11.         select a node according to the probability
12.         if ant.load + node.load > ant.capacity then
13.           return to the depot and go to 4
14.         else
15.           visit the selected node
16.     return to the depot
17.   save the best solution if found
18.   evaporate pheromone trails
19.   update pheromone trails
20. return the best solution
```

Figure 1: ACO algorithm in pseudo code.

In point 16, after visiting all nodes, each ant returns to its colony. Then, if the best solution found in the generation is better than the best solution found in previous ones, it is saved (see point 17). Point 18 ensures evaporating the pheromone trails and in point 19, pheromone trails are updated according to the given method. Then, the next generation begins until the termination condition is met. The best solution found is returned at the end of the algorithm in point 20.

3.2 Parameters

The algorithm works with a lot of parameters influencing the problem solution. This section presents most of them. Details are however beyond the scope of this paper. Some of these parameters are adopted from related problems, some of them were proposed by the authors as new. The parameters are as follows:

- Optimization type;
- Number of ants in colonies;
- Method of selecting depots;
- Pheromone evaporation coefficient;
- Method of updating pheromone trails;
- Pheromone updating coefficient;
- Coefficients to control the influence of pheromone trails, distance between nodes, distance between the node and depot, load in the node, and time interval when computing probabilities for visiting individual nodes;
- Termination conditions.

Optimization type allows selecting the optimization criterion (minimization of total distance travelled, or total time of the whole process taken, or total fuel consumed).

Number of ants in colonies n_a (see point 2 in Figure 1) is a parameter determining the number of different solutions to be created and evaluated within a generation. Pheromone trails are then updated according to these solutions (based on the selected method).

Method of selecting depots (see point 7 in Figure 1) is a key factor. We proposed five possibilities as follows:

- *Random selection*: depot (i.e. its vehicle) is selected randomly.
- *Selection of an idle depot*: depot with the shortest distance traveled so far (or time or fuel according to the given optimization criterion) is selected.
- *Selection of an idle depot (probability model)*: selection probabilities for all depots are computed based on the distance traveled so far (i.e. depots with shorter routes are more likely to be selected).
- *Selection of a depot with the greatest potential*: depot with the greatest potential (i.e. with the greatest sum of pheromone trails to unvisited customers) is selected.
- *Selection of a depot with the greatest potential (probability model)*: selection probabilities for all depots are computed based on the sum of pheromone trails to unvisited customers (i.e. depots with the bigger sum are more likely to be selected).

In point 8 in Figure 1, after a colony is chosen, the probabilities for an ant from this colony of choosing the path to the one of so far unvisited nodes are computed according to formula (2).

$$p_{ij}^k = \frac{{\tau_{ij}^k}^{\alpha} \cdot {\eta_{ij}}^{\beta} \cdot {\mu_{ij}^k}^{\gamma} \cdot {\kappa_{ij}^k}^{\delta} \cdot {\upsilon_{ij}^k}^{\varepsilon}}{\sum_{l \in S_u} {\tau_{il}^k}^{\alpha} \cdot {\eta_{il}}^{\beta} \cdot {\mu_{il}^k}^{\gamma} \cdot {\kappa_{il}^k}^{\delta} \cdot {\upsilon_{il}^k}^{\varepsilon}} \text{ for all } j \in S_u, \tag{2}$$

where p_{ij}^k is a probability for an ant from the colony k in a node i to visit a node j,
τ_{ij}^k is strength of a pheromone trail from the colony k between nodes i and j,
η_{ij} is a multiplicative inverse of the distance between nodes i and j ($\eta_{ij} = {d_{ij}}^{-1}$),
μ_{ij}^k is a so-called savings measure [17],
κ_{ij}^k is a measure for including the influence of ant's current load [18],
υ_{ij}^k is a measure for including the influence of a time interval of node j,
$\alpha, \beta, \gamma, \delta, \varepsilon$ are coefficients controlling the influence of $\tau_{ij}^k, \eta_{ij}, \mu_{ij}^k, \kappa_{ij}^k, \upsilon_{ij}^k$,
S_u is a set of so far unvisited nodes in a solution ($S_u \subset S$).

The measure υ_{ij}^k represents the suitability of a vehicle k currently being in node i to go to node j within its given time interval. If it is not possible, υ_{ij}^k is set to zero, and so is the final probability p_{ij}^k. On the other hand, the value of this measure is inversely proportional to the time the vehicle needs to wait for the node to be opened (if there is no waiting the value is 1).

Pheromone evaporation coefficient ρ determines the speed of evaporating pheromone trails at the end of each generation (see point 18 in Figure 1) according to formula (3).

$$\tau_{ij}^k = (1 - \rho) \cdot \tau_{ij}^k \text{ for all } i, j \in V, \tag{3}$$

where τ_{ij}^k is strength of a pheromone trail from the colony k between nodes i and j,
ρ is the pheromone evaporation coefficient.

Method of updating pheromone trails (see point 19 in Figure 1) is another key parameter. There are three possibilities as follows:

- *Update according to the best solution found in a current generation*: only the best solution found in a current generation influences the pheromone trails update process.
- *Update according to the best solution found so far*: only the best solution found so far influences the pheromone trails update process.
- *Update according to all solutions found in a current generation*: all solutions found in a current generation influences the pheromone trails update process.

Pheromone updating coefficient σ controls the influence of individual solutions when updating pheromone trails (see point 19 in Figure 1). The right ratio of the pheromone evaporation and updating coefficients is very important as bad values can cause the fast evaporation of pheromone trails on the one hand or the solution to get stuck in a false optimum on the other. Formula (4) presents the update of pheromone trails when updating according to the best solution found in a current generation.

$$\tau_{ij}^{k} = \tau_{ij}^{k} + x_{ij}^{k} \cdot \sigma \cdot \frac{d_{min}}{d_1} \quad \text{for all} \quad i, j \in V, \tag{4}$$

where τ_{ij}^{k} is strength of a pheromone trail from the colony k between nodes i and j,
x_{ij}^{k} is defined according to formula (1),
σ is the pheromone updating coefficient,
d_{min} is the distance of the best solution found so far,
d_1 is the distance of the best solution found in a generation.

Values of **coefficients $\alpha, \beta, \gamma, \delta, \varepsilon$** are also very important parameters of the algorithm as they control the influence of pheromone trails, distance between nodes, distance between the node and depot, load in the node, and time intervals of nodes when computing probabilities for visiting unvisited nodes.

The last set of parameters consists of several **termination conditions** which can be used (but does not have to) to control the total number of generations. These are as follows:

- Maximum number of generations.
- Maximum run time of the algorithm.
- Maximum number of generations without solution improvement.

4 EXPERIMENTS AND RESULTS

We carried out a great amount of experiments to find the best values for parameters mentioned in the previous section. It is however beyond the scope of this article to discuss them in detail. This section provides experiments conducted on benchmark problems for which we choose Cordeau's MDVRP-TW instances taken from [3].

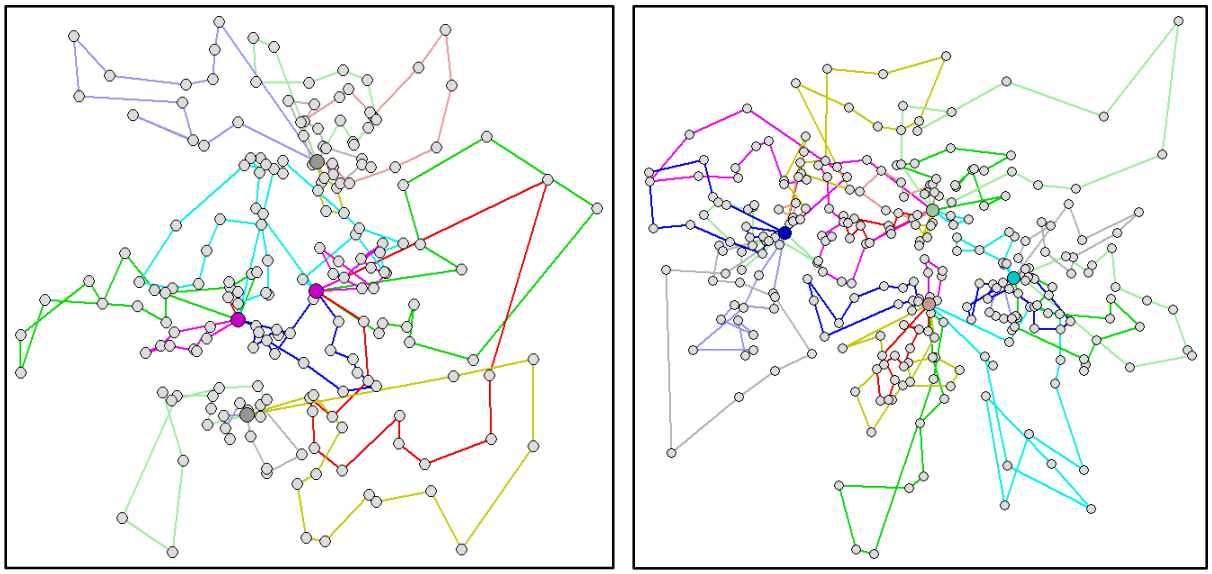
Table 1 shows the results for 6 benchmark instances, namely pr01, pr02, pr03, pr04, pr05, and pr06. As shown in Table 1, these instances have a wide range of complexity (from 48 to 288 nodes). We conducted 100 tests on each instance and recorded the best solution found, the mean and standard deviation. The last column shows the difference from the best known solution (taken form [3]).

Benchmark instance	Number of customers	Number of depots	Best known solution	Our best solution	Mean	Standard deviation	Error
pr01	48	4	1083.98	1092.38	1114.98	14.34	0.77%
pr02	96	4	1763.07	1865.97	1893.92	9.09	5.84%
pr03	144	4	2408.42	2554.22	2598.69	17.05	6.05%
pr04	192	4	2958.23	3025.98	3078.35	27.65	2.29%
pr05	240	4	3134.04	3372.52	3419.52	35.71	7.61%
pr06	288	4	3904.07	3957.48	4043.65	39.72	1.37%

Table 1: Results for MDVRP-TW benchmark problems.

As we can see, no solution found by our algorithm is worse than 8% compared with the best solution known so far. In case of pr06 which is a very complex problem (N = 288), the solution is very close to the best known result (difference only 1.37%).

Figure 2 presents two of the best solutions from Table 1 (namely pr04 and pr06) to illustrate the algorithm functionality to find high-quality solutions. All examples are metric (Euclidean) capacitated MDVRP-TW problems.



a) N = 192, M = 4 b) N = 288, M = 4

Figure 2: Experiments with the ACO algorithm on complex problems.

5 CONCLUSIONS

Ant colony optimization algorithm we proposed for MDVRP-TW problems was based on our version for MDVRP problems without time windows. As the previous section shows, the algorithm is able to find high-quality solutions even for complex instances. The algorithm ensures fast convergence close to the optimal solution.

The strength of the algorithm consists in its ability to be executed in parallel as the process of finding the solution by individual ants in colonies (see points 2 to 16 in Figure 1) is independent on one another. In the current version we distributed the computation to the individual cores of a multi-core processor, and thus shortened the whole run time several times.

Another advantage is the universality of the algorithm. It works not only with metric problems as shown in Section 4 but also with more universal cases: non-metric and even asymmetric

problems. It is caused by the basic principle of the algorithm, i.e. the assumption of triangle inequality is not applied here.

The perspectives of our future work consist in distribution of some computation to a GPU processor and also in transformation of parallel processing not only to the cores of a multi-core processor but also among more computers (e.g. within GRID networks).

The previous version of our algorithm (without time windows) has already been implemented in our tactical information system which supports commanders in their decision making processes. It is being used for planning the optimal supply distribution from multiple depots to friendly units on the battlefield. We plan to replace this solution by the advanced version which supports time windows for each customer.

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